

Tax Assessment Platform

What the platform does, what has been built, what is planned next, and how a new jurisdiction is added.

At a glance. A configuration-driven platform that carries a tax assessment from initiation to closure — evidence, calculation with full working, review and approval by different people, legal notice, objection, appeal, reassessment and closure. Built end to end and tested through the real screens; the United Kingdom and Saudi Arabia are configured; production readiness, independent assurance and integrations are the next phases.

Audience	Senior leadership and clients
Version	1.0
Date	30 September 2026
Status basis	The platform as built on branch <i>feat/documentated-e2e-cycle</i> , checked against its code, tests and a running instance
Classification	Confidential — for the intended recipients

Important. The tax rates, bands, penalties, interest rates and statutory periods configured for the United Kingdom and Saudi Arabia are illustrative. They are drawn from public guidance and have not yet been certified by a tax subject-matter expert. They demonstrate how the platform works, not a legal position.

Contents

1	Executive summary
2	What the platform does
3	Who uses it, and the checks between them
4	A tax assessment case from start to finish — the flow diagrams
5	Every possible case journey
6	Complete reference: every step a case can take
7	What has been delivered
8	What is partly delivered
9	Roadmap: what is pending, in phases
10	Jurisdictions: today, and how a new one is added
11	Safeguards built into the platform
12	Architecture at a glance
13	Quality and assurance
14	Connections to other systems
15	Known limitations and risks
16	Glossary

1. Executive summary

The Tax Assessment Platform is software a tax authority uses to assess what a taxpayer owes and to carry that assessment through to its end. It covers the whole life of an assessment: choosing who to examine, gathering the evidence, working out the adjustments, calculating the liability, having it reviewed and approved by different people, issuing and serving the legal notice, handling the taxpayer's objection and any appeal, reassessing where a decision requires it, and finally settling, closing, writing off or time-barring the case.

It is built so that **each country's tax rules are configuration, not software**. Rates, bands, deadlines, notice wording, approval limits, reference lists, calendars and risk rules are data. The United Kingdom (Corporation Tax) is the first jurisdiction; Saudi Arabia (Corporate Income Tax) has been configured alongside it to prove that a second country can be added without rewriting the platform.



Where the platform stands

- **Delivered:** the complete assessment lifecycle, working end to end through the screens; exact money arithmetic with a step-by-step calculation trace; separation of duties between preparer, reviewer and approver; notices with tamper-evidence and public verification; statutory deadlines on each country's working calendar; objections, appeals, reassessment and closure; a taxpayer portal; dashboards, eight reports and exports; risk-based selection; a visual form builder; and a process modeller.
- **Partly delivered:** approval by amount band is routed and recorded, but the approver's level is not yet enforced at the approval step; multi-factor sign-in is configured but not switched on; the connections to the authority's own filing, payment and registry systems use the platform's internal data today.
- **Pending:** production readiness and independent assurance (tax-expert sign-off, user acceptance testing, penetration test, accessibility audit, performance testing, deployment packaging), digital signing of notices, Arabic-language content, self-service configuration screens, and integrations with external systems. Section 9 proposes how these are phased.

The one rule the platform is built around: forms display, the server decides. Nothing that determines a legal figure or a statutory date is ever computed in a browser; every figure can be reproduced and explained, line by line, years later.

2. What the platform does

An assessment moves through fourteen stages. The table describes each in plain terms; section 4 shows every path between them.

#	Stage	What happens
1	Initiation	The authority decides to examine a taxpayer, a tax and a period — by a supervisor's decision, from a risk-based selection run, or automatically. One open case per taxpayer, tax and year is enforced.
2	Evidence retrieval	The platform gathers the filed return and the taxpayer's account (payments, credits, losses), then freezes that evidence with a digital fingerprint so the facts a decision rested on can later be shown to be unchanged. If a mandatory source cannot answer, the case waits rather than being assessed on nothing.
3	Assignment	A supervisor assigns the case to a named assessor. The case receives its reference number.
4	Preparation	The assessor records adjustments to what the taxpayer declared — each with a type, a reason and a narrative — and can ask the taxpayer for information, which pauses the case until a reply or the due date.
5	Calculation	The server calculates the liability through a fixed sequence: assessed base, loss set-off, taxable base, tax at the applicable rates, surcharge, credits, penalties, interest and net position. Every step is stored with its working.
6	Review	A different officer reviews the work. They accept it or return it for rework with a written reason. The preparer can never review their own work.
7	Approval	The case is routed to an approver according to the amount at stake, never by choice. The approver approves or rejects with a reason. While a case is under review or approval its figures are locked, so the approver approves exactly what the reviewer saw.
8	Finalisation	The figure becomes the legal determination. Losses the calculation relied on are consumed, so the same loss can never relieve two assessments.
9	Notice	The assessment notice is produced from approved wording (which can insert values but never calculate), stored as a PDF and given a fingerprint.
10	Service	The notice is served by e-mail, portal, post, hand delivery or publication. The legal "deemed served" date is worked out from each country's rules, and the taxpayer's statutory window opens.
—	Taxpayer's response	The taxpayer pays (the platform concludes this from the account, nobody can simply mark it paid), objects, or lets the window lapse. A supervisor may write off an uncollectable liability, with a reason.
11	Objection	The objection is recorded (even if late — lateness is measured, not used to refuse), its admissibility decided, any deposit assessed, committee opinions recorded, and the objection officer decides: allowed, partly allowed or rejected.
12	Appeal	A rejected objection can go to an external forum. Officers record the hearing and outcome — upheld, varied, set aside or remanded — and whether the outcome has been implemented.
13	Reassessment	Where a decision changes the figure, the case is reworked and goes through review and approval again. New information after a case has ended opens a linked successor case.
14	Closure	The case is closed with its final balance frozen and a retention period set; it may be placed under legal hold. Other endings are cancellation, write-off and time-bar.

3. Who uses it, and the checks between them

Role	What they do
Supervisor	Opens and assigns cases, runs risk selection, records paper objections and appeals, records payments, reassesses, places legal holds, closes, cancels and writes off (cancelling and writing off always need a written reason).
Assessor	Prepares the case: adjustments, information requests, calculation, submission for review.
Specialist	May record adjustments on a case (for example transfer pricing).
Reviewer	Accepts a prepared case or returns it for rework, with a reason.
Approver (levels 1–3)	Approves or rejects cases in their amount band; rejection needs a reason.
Notice issuer	Issues and serves notices for the office.
Objection officer	Records objections, rules on admissibility and decides them — the only role that may decide.
Committee member	Records opinions on objections they are convened for.
Appeals officer	Records appeals, hearings, outcomes and their implementation.
Auditor (read-only)	Reads the register, case histories, calculations and configuration; changes nothing.
Administrator	Maintains configuration: publishes rule sets and notice wording, deploys process definitions, manages the permission catalogue.
Taxpayer	Uses the portal to see their own served assessments, notices and account, and to file objections.
System	Carries out automatic steps: evidence retrieval, approval routing, finalisation, notice generation, response windows, settlement and time-bar.

Checks between people (separation of duties)

- The person who prepared a case may not review it; the preparer and the reviewer may not approve it. This is checked against who actually acted on the case, not against who currently holds it.
- Routing to an approver and finalisation are automatic, so nobody can steer their own case to a friendlier approver or finalise an unapproved one.
- Whoever writes a rule set cannot publish it; a second person must.
- Only the objection officer decides an objection; a supervisor cannot.
- A taxpayer only ever sees their own affairs. Which taxpayer a login acts for is read from recorded, dated authority on every request, never from anything the browser sends.
- Every screen offers only the actions the person may take at that point, and the server checks every action again regardless.

4. A tax assessment case from start to finish

The overview shows the seven broad phases a case passes through and the loops between them. The two detailed diagrams that follow show **every** status a case can be in and **every** step between them — all `TRANSITION_COUNT` defined steps, plus the successor case opened on new information. Nothing a case can do is left out: the diagrams are drawn from the same definition the platform enforces.

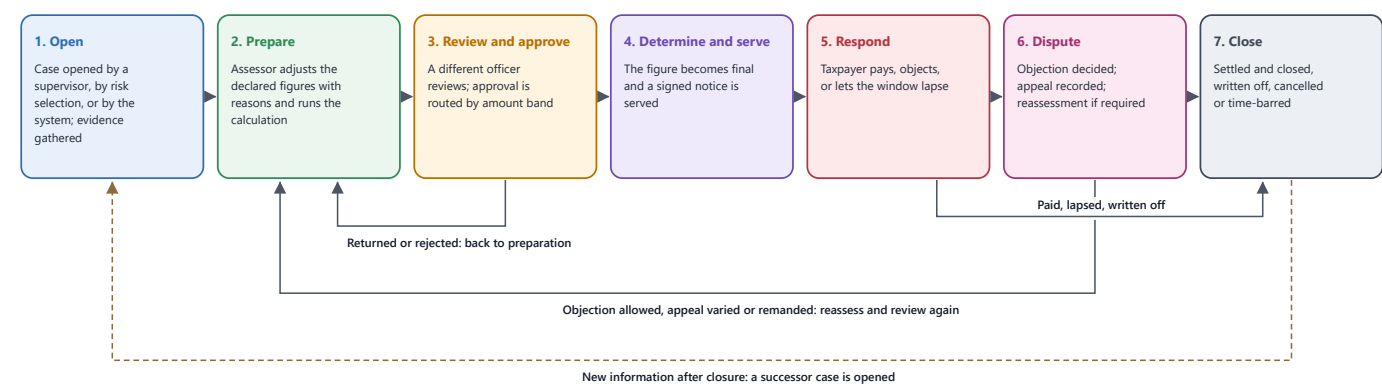


Figure 1 — The phases of a case, with the loops back to preparation and the successor case.

How to read the detailed diagrams

- Opening
- Preparation and rework
- Review and approval
- Determination and service
- Objection
- Appeal
- Endings (thick border = final)

- Each arrow names the step and, in brackets, who may take it.
- * means a written reason is mandatory and is recorded permanently with the step.
- A dashed arrow opens a *new* case linked to the old one. ⤵ marks the endings from which that can happen.
- Dark tabs connect Part 1 and Part 2.
- Two further statuses, *Finalisation failed* and *Notice failed*, are reserved for future error handling; no step leads to them today.

Figure 2 — Complete case flow, Part 1: opening to the served notice

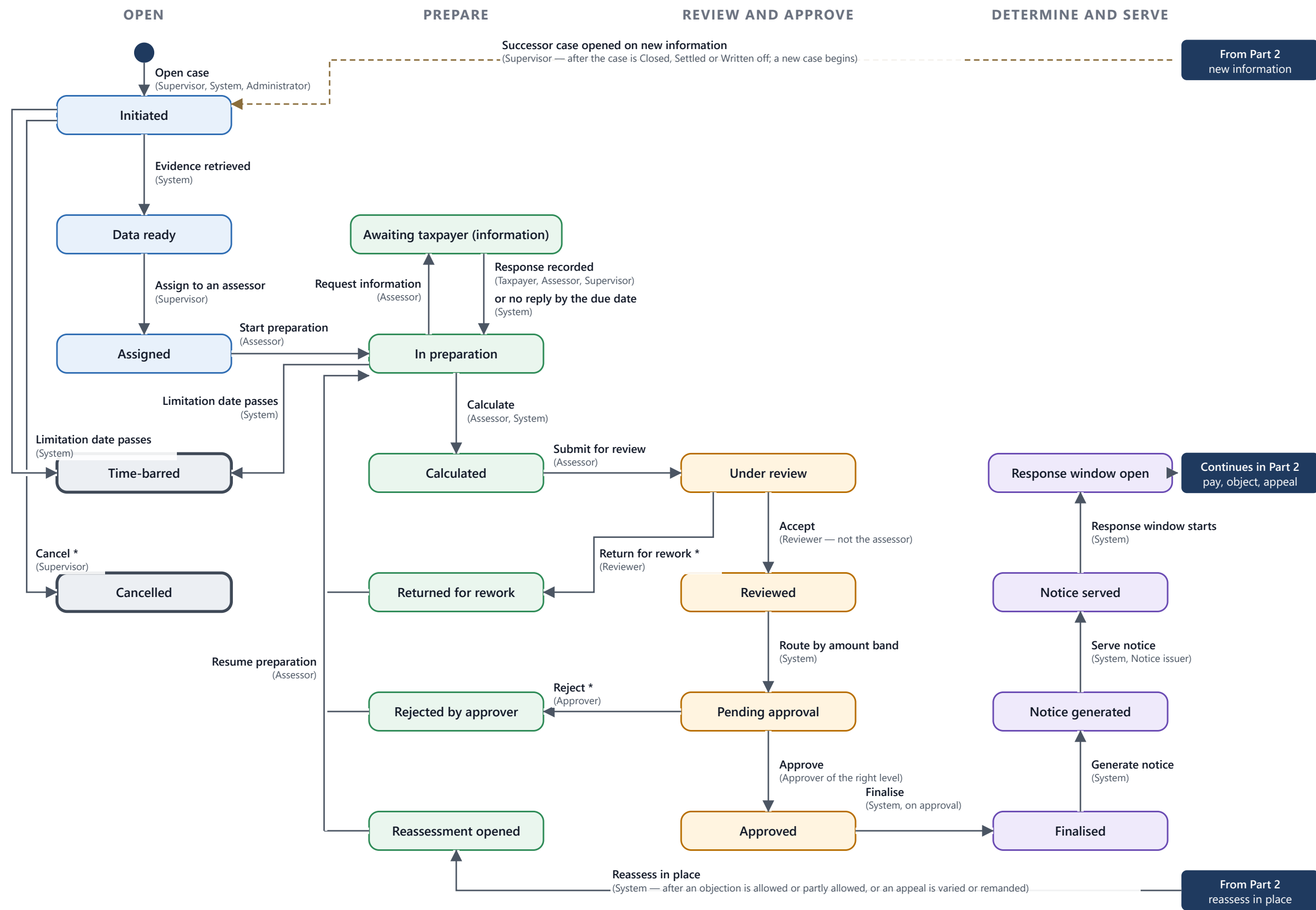
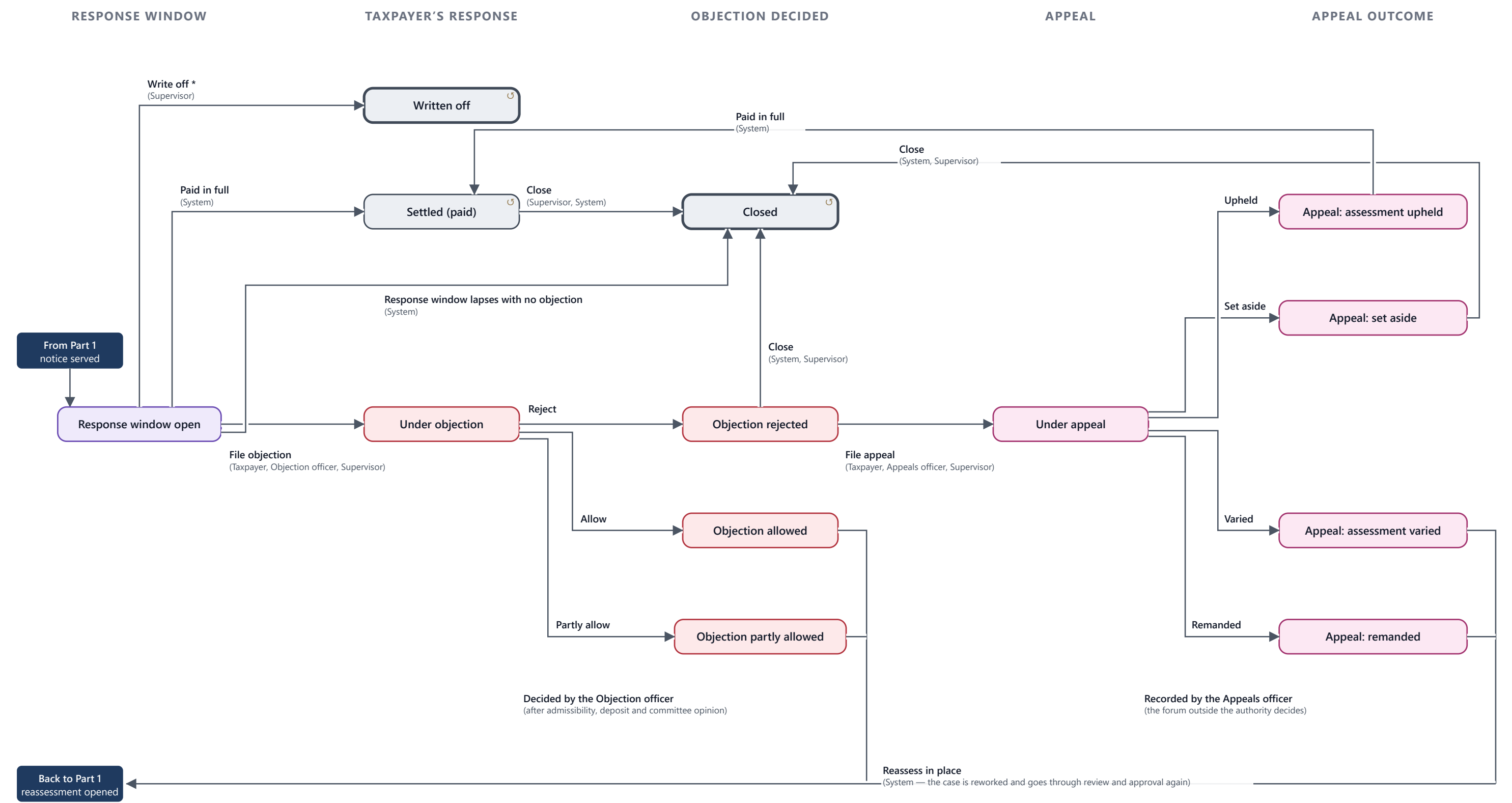


Figure 3 — Complete case flow, Part 2: after the notice



5. Every possible case journey

These are the distinct ways a case can travel through the platform. Several can combine: for example, a case can be returned for rework twice, then approved, served, objected to and reassessed.

1. Straightforward assessment, paid. Opened, evidence retrieved, assigned, prepared, calculated, reviewed, approved, finalised, notice served; the taxpayer pays in full; the case is settled and closed.

Initiated → Data ready → Assigned → In preparation → Calculated → Under review → Reviewed → Pending approval → Approved → Finalised → Notice generated → Notice served → Response window open → Settled → Closed

2. No response from the taxpayer. As above, but the response window lapses with no objection; the case closes automatically.

... → Response window open → Closed

3. Information requested during preparation. The assessor asks the taxpayer for information; the case waits until the response is recorded or the due date passes, then preparation continues.

In preparation → Awaiting taxpayer → In preparation

4. Returned by the reviewer. The reviewer returns the case with a reason; the assessor reworks it and resubmits.

Under review → Returned for rework → In preparation → ... → Under review

5. Rejected by the approver. The approver rejects with a reason; the case goes back to preparation and through review and approval again.

Pending approval → Rejected by approver → In preparation → ...

6. Cancelled before work begins. A supervisor cancels a newly opened case, with a reason (for example, opened against the wrong period).

Initiated → Cancelled

7. Time-barred. The legal time limit for assessing passes while the case is still being opened or prepared. Automatic time-bar is off by default and never applies under legal hold.

Initiated or In preparation → Time-barred

8. Written off. After the notice, a supervisor writes off a liability that cannot be collected, with a reason.

Response window open → Written off

9. Objection allowed. The taxpayer objects; the objection officer allows it; the case is reassessed and goes through review and approval again before a new figure is served.

Response window open → Under objection → Objection allowed → Reassessment opened → In preparation → ...

10. Objection partly allowed. As journey 9, for a partial change.

Under objection → Objection partly allowed → Reassessment opened → ...

11. Objection rejected, no appeal. The objection is rejected and the case is closed.

Under objection → Objection rejected → Closed

12. Appeal: assessment upheld. The taxpayer appeals; the forum upholds the assessment; the taxpayer pays; the case settles and closes.

Objection rejected → Under appeal → Appeal upheld → Settled → Closed

13. Appeal: assessment varied. The forum changes the figure; the case is reassessed to match and reviewed and approved again.

Under appeal → Appeal varied → Reassessment opened → ...

14. Appeal: remanded. The forum sends the case back for a fresh determination; it is reassessed.

Under appeal → Appeal remanded → Reassessment opened → ...

15. Appeal: set aside. The forum sets the assessment aside; the case is closed.

Under appeal → Appeal set aside → Closed

16. New information after the case has ended. A closed, settled or written-off case is reassessed on new information by opening a successor case, linked to the original, which starts again at the beginning. Reassessing beyond the legal time limit needs a written reason and a named authoriser.

Closed / Settled / Written off → new case: Initiated → ...

17. Evidence source unavailable. If a mandatory evidence source cannot answer, the case stays Initiated until it can, rather than being assessed on incomplete facts. It can still be cancelled or time-barred from there.

Initiated (waits) → Data ready

18. Legal hold. At any point a supervisor can place a case under legal hold, with a reason. The hold does not change the status; it blocks automatic time-bar and retention-driven disposal until lifted.

6. Complete reference: every step a case can take

The platform's own definition of the case lifecycle, in full. A step not in this table cannot happen; a person not listed against a step cannot take it; where "Reason" is marked, the step is refused without a written reason, and the reason is kept permanently in the case history.

#	From	Step	To	Who may take it	Reason
1	— (start)	Open the case	Initiated	Supervisor, System, Administrator	
2	Initiated	Evidence retrieved from every mandatory source	Data ready	System	
3	Initiated	Cancel the case	Cancelled	Supervisor	Yes
4	Data ready	Assign to a named assessor	Assigned	Supervisor	
5	Assigned	Start or resume preparation	In preparation	Assessor	
6	In preparation	Request information from the taxpayer	Awaiting taxpayer (information)	Assessor	
7	Awaiting taxpayer (information)	Record the taxpayer's response	In preparation	Taxpayer, Assessor, Supervisor	
8	Awaiting taxpayer (information)	No reply by the due date	In preparation	System	
9	In preparation	Calculate the liability	Calculated	Assessor, System	
10	Calculated	Submit for review	Under review	Assessor	
11	Under review	Return for rework	Returned for rework	Reviewer	Yes
12	Returned for rework	Start or resume preparation	In preparation	Assessor	
13	Under review	Accept the prepared assessment	Reviewed	Reviewer	
14	Reviewed	Route to the approver band for the amount	Pending approval	System	
15	Pending approval	Approve	Approved	Approver L1, Approver L2, Approver L3	
16	Pending approval	Reject	Rejected by approver	Approver L1, Approver L2, Approver L3	Yes
17	Rejected by approver	Start or resume preparation	In preparation	Assessor	
18	Approved	Finalise: the figure becomes the determination	Finalised	System	
19	Finalised	Generate the assessment notice	Notice generated	System	
20	Notice generated	Serve the notice	Notice served	System, Notice issuer	
21	Notice served	Open the taxpayer's response window	Response window open	System	
22	Response window open	Paid in full (derived from the account)	Settled (paid)	System	
23	Response window open	File an objection	Under objection	Taxpayer, Objection officer, Supervisor	

24	Response window open	Response window lapses with no objection	Closed	System	
25	Under objection	Allow the objection	Objection allowed	Objection officer	
26	Under objection	Partly allow the objection	Objection partly allowed	Objection officer	
27	Under objection	Reject the objection	Objection rejected	Objection officer	
28	Objection allowed	Reassess in place	Reassessment opened	System	
29	Objection partly allowed	Reassess in place	Reassessment opened	System	
30	Objection rejected	File an appeal	Under appeal	Taxpayer, Appeals officer, Supervisor	
31	Objection rejected	Close the case	Closed	System, Supervisor	
32	Under appeal	Record: forum upheld the assessment	Appeal: assessment upheld	Appeals officer	
33	Under appeal	Record: forum varied the assessment	Appeal: assessment varied	Appeals officer	
34	Under appeal	Record: forum set the assessment aside	Appeal: set aside	Appeals officer	
35	Under appeal	Record: forum remanded the case	Appeal: remanded	Appeals officer	
36	Appeal: assessment varied	Reassess in place	Reassessment opened	System	
37	Appeal: remanded	Reassess in place	Reassessment opened	System	
38	Appeal: set aside	Close the case	Closed	System, Supervisor	
39	Appeal: assessment upheld	Paid in full (derived from the account)	Settled (paid)	System	
40	Reassessment opened	Start or resume preparation	In preparation	Assessor	
41	Settled (paid)	Close the case	Closed	Supervisor, System	
42	Initiated	Limitation date passes	Time-barred	System	
43	In preparation	Limitation date passes	Time-barred	System	
44	Response window open	Write off the liability	Written off	Supervisor	Yes
45	Closed	Reassess on new information: a successor case is opened	Initiated (new case)	Supervisor	Yes
46	Settled (paid)	Reassess on new information: a successor case is opened	Initiated (new case)	Supervisor	Yes
47	Written off	Reassess on new information: a successor case is opened	Initiated (new case)	Supervisor	Yes

Rows 45–47 are the successor-case paths: they open a new, linked case rather than moving the existing one. Payment ("paid in full") is never a button: the platform concludes it from the taxpayer's account.

7. What has been delivered

Everything in this section is built, connected to the screens, and covered by automated tests or a documented run against a working instance.

Area	What a user can do
Case register and work queues	Search, sort and page through every case they may see (the list is limited by role on the server); seven work queues by stage with counts; export the filtered register to CSV or Excel, with each export recorded and limited.
Case workbench	One screen per case with tabs for evidence, adjustments, calculation, deadlines and service levels, notices, disputes, closure, the process journey and the full history. It offers only the actions the person may take at that moment.
Evidence	Retrieve evidence from every configured source in one action; see the declared figures and each source's fingerprint.
Adjustments	Record adjustments through a configurable form; amounts are handled as exact text, never as approximate numbers.
Calculation	Calculate on the server; see every step of the working; compare versions to see what changed; preview a "what if".
Review and approval	Accept or return; route by amount band with a sentence explaining which band was chosen and why; approve or reject.
Notices	Issue the assessment notice from approved wording; download the PDF; serve it by any configured channel with proof of each attempt; verify any notice's integrity — officers internally, anyone publicly by reference.
Deadlines and service levels	Statutory deadlines computed on each country's working calendar, each with a plain sentence explaining how it was derived; internal service-level clocks shown separately.
Objections	File (including late, with lateness measured), rule on admissibility, assess the deposit where the country requires one, record committee opinions, decide.
Appeals	File to a forum recognised by the country, record hearings and the outcome, and track whether the outcome has been implemented.
Reassessment	Reassess in place after a dispute outcome, or open a linked successor case after closure; see every assessment of the period.
Payments, settlement and closure	Record payments against the taxpayer's account (a bank reference can only be credited once); settlement is concluded from the figures; close with a reason and retention period; legal hold; write-off with a reason; automatic nightly closure of settled cases.
Risk-based selection	Score taxpayers against the country's weighted risk rules, see which rules fired, and open cases from a run as a separate, capped act.
Taxpayer portal	A taxpayer sees their own served assessments in plain language, their notices, what they owe and their objections, and can file an objection.
Dashboard and reports	A dashboard scoped to the viewer, with totals shown per currency; seven management reports (assessment summary, collection, adjustment analysis, dispute outcomes, ageing, deadline exposure, unserved notices) and a reconciliation report.
Forms	A visual form builder with live preview and publishing; the same form engine validates in the browser and again on the server.
Process coordination	A workflow engine coordinates who is asked to act and when, with task inboxes and timers; a process modeller lets an administrator check and deploy a process definition; each case shows where it has reached.

Security and audit	Single sign-on; a permission catalogue in which every route must be listed or it is refused; a permanent, tamper-resistant case history; redacted access logs; document access history.
Languages and themes	The screens are built on display keys for translation, with right-to-left support; light and dark themes.
Operations	Six scheduled jobs (notifications, retries, exports, deadline sweep, auto-closure, workflow reconciliation) that are safe to run on several servers; a security boundary probe; a disaster-recovery rehearsal that restores the database and proves every figure matches.

8. What is partly delivered

These work, but with a stated limit. Each limit is also scheduled in the roadmap.

Capability	What works	The limit today
Approval by amount band	The case is routed to the band for its amount; the band and its rule are recorded.	The approval step accepts any approver level; a band's required level and a second approval at level 3 are not yet enforced.
Multi-factor sign-in	Configured for every officer account.	Not switched on; an environment decision.
Evidence sources	Filed return and taxpayer account, through replaceable interfaces.	Both read the platform's own tables; no external filing system, bank feed or registry is connected yet.
Notifications	E-mail and in-portal messages, queued, retried and recorded.	No SMS provider; templates in English only.
Rule sets	Versioned, date-effective, dual-control publishing, and a simulator that replays a draft over past cases.	New rule sets are loaded as data by the delivery team; there is no editing screen yet.
Configurable forms	The adjustment form is fully configurable.	The other assessment forms (including objection and appeal) are built into the screens.
Process coordination	The engine coordinates from opening up to the objection being filed.	Objection, appeal and reassessment are worked through the screens without engine coordination; review and approval escalation timers are not modelled.
Taxpayer portal	Assessments, notices, account, objections.	No appeal filing, payment or reply to information requests yet.
Languages	English throughout; the Arabic language and right-to-left layout exist.	No Arabic labels or notice wording yet; Arabic PDFs are refused rather than rendered blank.
Administration	View the permission catalogue, approval bands and scheduled jobs.	Users, roles, reference lists and labels cannot yet be edited on screen.
Time-bar	Built and bounded by strict rules.	Off by default; no jurisdiction yet configures the limitation deadline it needs.

9. Roadmap: what is pending, in phases

The platform's plan describes phases 0 to 8, which have been built with the exceptions listed here. The phasing below is **a proposal for sequencing the remaining work**, grouped so that each phase delivers something usable. It is to be agreed with leadership and the first client; order and scope may change with their priorities.

Phase 9 — Production readiness and independent assurance

- Tax subject-matter expert sign-off of every rule, rate and deadline, and of the golden calculation tests.
- User acceptance testing with tax officers (the 12-scenario pack is prepared).
- Independent penetration test; accessibility audit to WCAG 2.1 AA; performance testing of calculation, notice generation and concurrent users.
- Deployment packaging (container images and orchestration), environment promotion, monitoring and metrics, operational runbooks; disaster recovery extended to document storage, identity and in-flight workflow timers.
- End-to-end browser tests and the workflow engine's tests run automatically on every change.
- Multi-factor sign-in switched on; captcha on public verification.

Phase 10 — Control completeness

- Enforce the approver level required by each amount band, and the second approval at the highest level.
- Write-off authority limits; assessor override of a calculated figure, with justification and approval.
- A rule-set publisher may not work cases computed under that rule set.
- Escalation timers for overdue review and approval; time-boxed delegation of authority.
- Configurable "deemed allowed" objections and late-filing condonation; a disposal list for cases past retention.

Phase 11 — Connections to the authority's systems

- Filed-return system, payment ledger and revenue accounting, taxpayer registry — each through its existing interface, so the tax logic does not change.
- Digital signature and verification QR code on notices; SMS gateway.
- Currency conversion where a jurisdiction needs it.

Phase 12 — Configuration self-service

- Screens to author and version rule sets (rates, bands, thresholds, penalties, interest), deadlines, and reference lists and labels; user and role administration.
- All eighteen assessment forms as configurable templates, including jurisdiction-specific objection and appeal forms.
- Configuration promoted between environments as reviewed files, with two-person approval.

Phase 13 — Taxpayer experience and language

- Portal: appeal filing, payment, replies to information requests.
- Arabic labels and notice wording, with right-to-left PDF notices.
- Engine coordination of the objection and appeal stages.

Phase 14 — Breadth

- Further tax types — VAT, withholding tax, personal income tax, excise — and individual taxpayers.
- Risk-model feedback (selection measured against assessment yield), officer productivity and further analytics.
- Further jurisdictions, using the method in section 10.

10. Jurisdictions: today, and how a new one is added

Configured today

Both jurisdictions assess corporate income tax for companies. The United Kingdom is the first jurisdiction; Saudi Arabia was configured to prove that a second country is data, not a rewrite.

Configuration	United Kingdom (GB) — Corporation Tax	Saudi Arabia (SA) — Corporate Income Tax
Currency	Pound sterling (GBP)	Saudi riyal (SAR)
Rule set	GB-CIT-FY2024, effective 1 April 2023	SA-CIT-2024, effective 1 January 2024
Rates	19% up to 50,000; 25% main rate to 250,000 with marginal relief	Flat 20%
Losses and credits	Losses oldest first; credits: double tax relief, withholding, advance payment	Losses oldest first; credits: withholding, advance payment, double tax relief
Penalties	Late filing: 100 on day 1, 100 more at 90 days, 10% at 180 days	Late filing 1% (capped); late payment 5% after 30 days (capped)
Interest	7.75% simple, 365-day year	5% simple, 360-day year
Deadlines	Filing 12 months after period end; payment 9 months and a day; objection 30 days from service; appeal 30 days from decision	Filing and payment 120 days after period end; objection 60 days from service; appeal 30 days from decision; information 20 working days
Working week	Monday–Friday; UK holidays	Sunday–Thursday; Saudi holidays
Deemed service	Registered post 2 working days; publication 7 days	Registered post 5 working days; publication 14 days
Approval bands	Up to 100k: level 1; to 1m: level 2; above: level 3 (two approvals)	Up to 500k: level 1; to 5m: level 2; above: level 3 (two approvals)
Objection deposit	None	10% of the disputed amount (minimum 1,000, maximum 500,000)
Appeal forums	First-tier Tribunal, Upper Tribunal, Court of Appeal	General Secretariat, Appellate Committee, Supreme Administrative Court
Retention	7 years / 20 years / permanent	10 years / 20 years / permanent
Risk rules	Non-filer, large loss, credit ratio, prior adjustment	Non-filer, prior adjustment
Notices	Assessment and demand, in English	Assessment, in English (Arabic pending)

All figures are illustrative and await tax-expert certification. Saudi Arabia has no golden calculation tests yet. For a production UK deployment, the UK reference lists, tax type and holidays (loaded today with the demonstration data) will be moved into the permanent configuration.

How a new jurisdiction is added

A new country is added by adding configuration: no change to the platform's code is needed for anything the platform already knows how to do. The steps are:

1. **Currency and tax type** — the currency and its decimal places; each tax type with its default currency.
2. **Working calendar** — the weekend days and the public holidays.
3. **Reference lists** — adjustment types and reasons, objection grounds, appeal forums, closure reasons and retention classes.
4. **Rule set** — rates and bands, thresholds, loss rules, credit order, penalties, interest and rounding, dated from when they apply; published by a second person.

5. **Statutory deadlines** — filing, payment, objection, appeal and information-response periods, each as "so many days or months from an event", on calendar or working days (and, if wanted, the limitation period).
6. **Deemed-service rules** — when a notice counts as served for each delivery channel.
7. **Approval bands** — the amount ranges and the approver level and number of approvals for each.
8. **Notice wording** — each notice type in each language.
9. **Objection deposit** — if the country requires one.
10. **Service levels and risk rules** — internal targets, and weighted selection rules.
11. **Notifications and labels** — messages and screen labels in each language.
12. **Forms** — the tax return and, where needed, adjusted assessment forms, built in the form builder.
13. **Taxpayers** — normally from the country's registry.
14. **Assurance** — a set of golden calculation tests signed by a local tax expert, and a simulation of the rule set over sample cases before it is published.

The process definition is shared and takes its durations from the country's deadlines, so it does not change.

When code is needed. Only when a country needs something the platform has never done: a new kind of calculation rule, a new risk indicator, a new deadline trigger or calendar rule, a new delivery channel (for example SMS), non-Latin PDF notices, currency conversion, or — until those forms become configurable — country-specific objection and appeal forms. Each is added once and is then available to every jurisdiction.

11. Safeguards built into the platform

Safeguard	What it means	Why it matters
Exact money	All amounts use exact decimal arithmetic and statutory rounding rules; ordinary computer arithmetic on money is blocked automatically. The money component has 100% test coverage.	A rounding error becomes a wrong liability on every case calculated after it.
Explainable calculation	The browser never calculates. Each result stores its full working, the rule-set version and a fingerprint of its inputs.	Any figure can be reproduced and explained in an appeal or in court.
Golden tests	42 complete cases pin inputs, results and working; a rule change must show an intended, reviewed difference.	A silent change to anyone's liability is caught before release.
Permanent case history	Every step writes one history entry in the same transaction; the database itself refuses edits or deletions of the history, even by an administrator of the database.	The record of who did what cannot be rewritten.
Fail-safe access	An action not in the permission catalogue is refused to everyone; an outage refuses rather than allows.	A missed setting or an outage never opens access.
Separation of duties	Preparer, reviewer and approver must be different people; figures are locked while under review and approval.	Four eyes on every legal figure; the approver approves what the reviewer saw.
Configuration cannot run code	Configuration chooses from a fixed list of behaviours; it never contains queries or formulas.	Administrative settings cannot become a route for attack.
Frozen evidence and determinations	Evidence is fingerprinted and frozen at submission; a finalised figure can only change by reassessment with grounds.	The facts and the figure a decision rested on stay provable.
Deadlines computed, never typed	Every statutory date is calculated from the recorded service date on the right calendar, with its derivation.	A wrong statutory window is legal exposure.
Bounded time-bar	Automatic time-bar is off by default and strictly limited when on; never under legal hold.	A time-bar ends a collectable debt and cannot be undone.
Notice integrity	Each notice has a fingerprint that officers and the public can verify.	A taxpayer or court can check a notice is unaltered.
Taxpayer isolation	A taxpayer's access is decided by recorded authority, never by anything sent from the browser.	No taxpayer can see another's affairs.
Controlled exports	Every export is recorded, scoped to the requester, rate-limited and protected against spreadsheet formula injection.	The register cannot be copied out anonymously.

12. Architecture at a glance

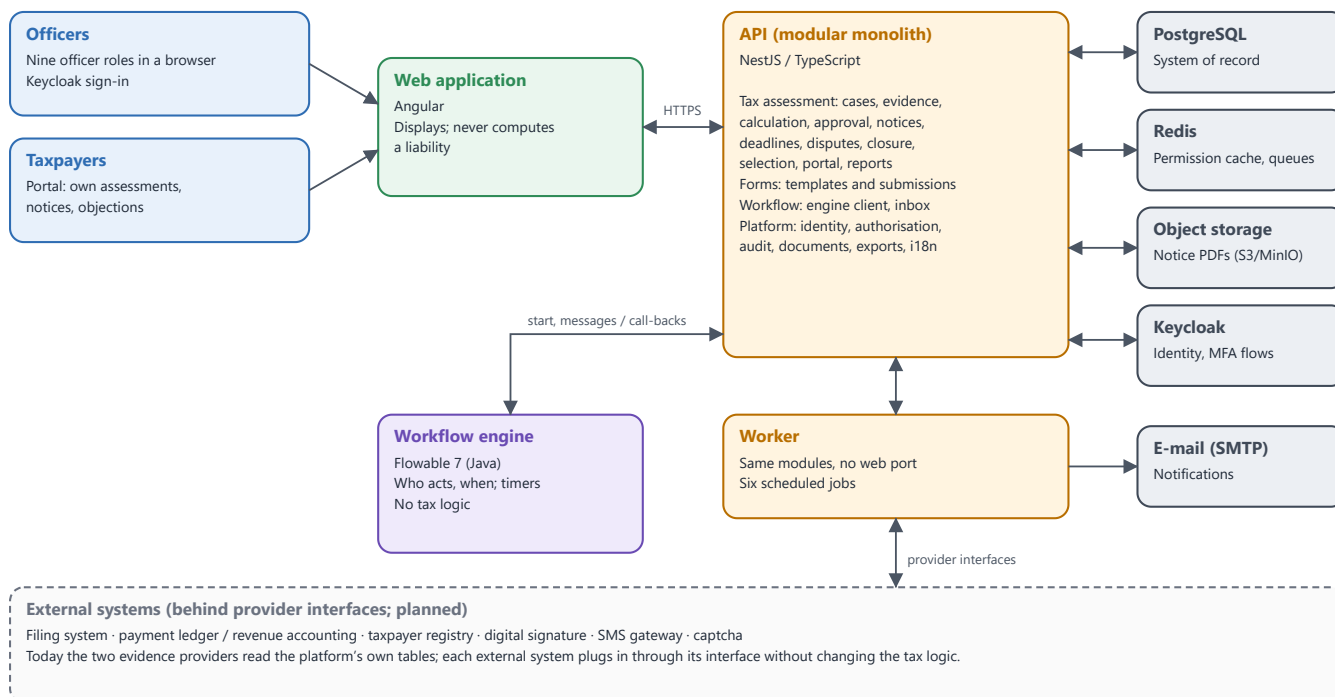


Figure 4 — The main components. External systems connect through interfaces, so replacing a provider does not touch the tax logic.

- **One language across the stack** (TypeScript) for the screens, the server and the form engine, so the same form rules run in the browser and on the server.
- **A modular monolith**: one server application with strictly separated modules (platform, forms, workflow, tax assessment). Modules can be split into separate services later if load or team structure requires it.
- **PostgreSQL** is the system of record; **Redis** holds the permission cache and queues; **Keycloak** provides identity; documents are held in **S3-compatible** storage.
- **Flowable** coordinates work but decides nothing about tax: the server alone decides a case's status. If the engine is unavailable, cases are worked by hand and a reconciliation job reports any difference.
- Designed to scale horizontally: the server and background worker can run as several copies behind a load balancer, with job locks so scheduled work runs once.

13. Quality and assurance

Suite	Tests	What it proves
Money component	74	Exact arithmetic and every rounding rule; 100% coverage, enforced on every change
Case lifecycle definition	31	The lifecycle is internally consistent: one start, no way out of a final status, every step has someone allowed to take it, every status reachable, figures locked under review
Form engine	108	The rules, formulas and validation shared by browser and server
Server (unit and integration)	313	Including the 42 golden calculation cases, deadlines, limitation, permissions, security properties, exports, the process validator
Screens (unit)	53	Screen components in a real browser
End to end	43	Real browsers driving the real screens against the real server, identity service and database: the full lifecycle through six officers, role separation, the register and exports, the taxpayer portal, the process modeller, and a documented full cycle. All 43 passed on 30 September 2026; the documented cycle exercised all 44 defined steps through the screens, with a screenshot of each. Each test is recorded as one paced video.

Every change is automatically checked for formatting, code rules (including the money and module-boundary rules), architectural guards, type safety, the database migrations applied, undone and re-applied, the test suites, the money coverage gate and known-vulnerable dependencies.

Also in place

- A security boundary probe of the running system: anonymous access, forged tokens, one taxpayer reaching another's data, role separation, injection, and rate limits. It automates a penetration tester's first checks; it is not a penetration test.
- A disaster-recovery rehearsal that backs up, restores and proves every figure, fingerprint and count matches.
- A load test at 920,000 cases and 200,000 taxpayers: the filtered register answered in 42 ms and the heaviest report in under half a second (measured once, by hand).

Not yet done

Tax-expert sign-off; user acceptance testing with officers; an independent penetration test; an accessibility audit; repeatable performance testing; end-to-end and workflow-engine tests running automatically on every change. All are in Phase 9.

14. Connections to other systems

System	Status	Detail
Identity (Keycloak)	In place	Single sign-on for officers and taxpayers; can federate to the authority's own directory (to be decided).
Workflow engine (Flowable)	In place	Coordinates tasks and timers; the definition is deployed by an administrator.
Document storage	In place	S3-compatible; every read is checked against its fingerprint; access is logged.
E-mail	In place	Through any SMTP relay.
Filed-return system	Internal	Reads returns filed through the platform's own forms; the external system connects through the same interface.
Payment ledger / revenue accounting	Internal	An internal account of payments, credits and losses; the largest scope decision for a client.
Taxpayer registry	Internal	Taxpayers held in the platform; the registry connection is planned.
Digital signature	Planned	Notices are fingerprinted and verifiable today; a signing service and QR code are planned.
SMS gateway	Planned	The channel exists; no provider is connected.
Courts and tribunals	Manual	Appeal outcomes are recorded by officers, by design in this version.

15. Known limitations and risks

Limitation	Consequence and mitigation
Tax content not yet certified	Rates and deadlines are illustrative. Mitigation: expert sign-off in Phase 9, before any live use; the golden tests make every change reviewable.
Approver level not enforced at approval	Any approver can approve a case routed to a higher band, although the band is recorded. Mitigation: Phase 10; until then, approval rights should be granted with this in mind.
Multi-factor sign-in not switched on	Configured but not active. Mitigation: switch on per environment in Phase 9.
Notices not digitally signed	Integrity rests on the fingerprint and verification service. Mitigation: signing service in Phase 11.
No production packaging yet	The backing services run in containers; the applications are not yet packaged for production. Mitigation: Phase 9.
External systems not connected	Evidence and payments come from internal data. Mitigation: Phase 11, through existing interfaces; the scope depends on each client's systems.
Arabic content missing	Arabic screens would show untranslated keys; Arabic notices are refused. Mitigation: Phase 13.
Performance measured once	Register performance is good at 920k cases; calculation and notice throughput are not yet measured. Mitigation: Phase 9 performance testing.

16. Glossary

Term	Meaning
Assessment / case	One assessment of one taxpayer, for one tax and one period.
Declared vs assessed	What the taxpayer filed, and what the authority determines.
Adjustment	A change from the declared to the assessed figure, with its type, reason, amount and explanation.
Evidence snapshot	The source data as retrieved, fingerprinted and never altered.
Rule set	A dated, versioned set of a country's tax rules used by the calculation.
Calculation trace	The stored, step-by-step working behind a figure.
Finalisation	The point at which an approved figure becomes the legal determination.
Notice / service	The legal document given to the taxpayer, and its delivery; statutory periods run from the service date.
Deemed service	A rule that treats a notice as served a set time after sending, without proof of receipt.
Objection	The taxpayer's first challenge to an assessment, decided within the authority.
Admissibility / condonation	Whether an objection will be heard; accepting one filed late.
Appeal / forum	A challenge decided outside the authority; the tribunal, court or committee that hears it.
Upheld / varied / set aside / remanded	The forum keeps the assessment; changes it; cancels it; sends it back for a fresh decision.
Reassessment	A revised assessment — in place after a dispute decision, or as a linked successor case after closure.
Limitation / time-bar	The legal time limit for assessing; a case that passes it is time-barred.
Settlement / write-off	The liability paid in full; an uncollectable liability abandoned with a reason.
Legal hold	A block on automatic time-bar and disposal while a matter is pending.
Retention	How long a closed case must be kept before it may be disposed of.
Separation of duties	The rule that preparer, reviewer and approver must be different people.
SLA vs statutory deadline	An internal service target, and a date set by law; shown separately.
Golden case	A complete test case whose expected result and working are fixed, so any change is visible.

Prepared from the platform's source code, tests, configuration, architecture decision records and a running instance, 30 September 2026.
Counts reflect the platform at that date.